



ADAMA SCIENCE AND TECHNOLOGY UNIVERSITY

School of Electrical Engineering and Computing Software Requirement Engineering (SEng3201)

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SRS – Internship Service Portal (SW and CSE)

CHAPTER ONE: PROJECT PROPOSAL

1.1 System Title

Internship Service Portal for the College of Electrical Engineering and Computing

1.2 Problem Statement

Internship placement and management is an important academic requirement for undergraduate students in the College of Electrical Engineering and Computing. However, the current internship process is mostly manual, slow, and inefficient, which creates many difficulties for students, internship coordinators, and the institution as a whole.

Currently, students must physically go to the department to collect internship request letters and then search for companies on their own. Once a company accepts them, students are required to return to the department and submit printed acceptance letters. For students who fail to find placements, internship coordinators manually contact companies by sending official letters and waiting for responses, which takes a lot of time and often causes delays.

Additionally, **ineligible students** such as non-batch students, graduates, and students outside 3rd and 4th year are able to register for internships due to the absence of a proper validation system. This creates unnecessary administrative workload and data inconsistency.

Further, the grading and assessment exercise is done by hand. This happens after the students have been evaluated based on company reports and interviews, where the grades are then determined manually and submitted physically to the registrar to be loaded. This takes time, is prone to errors, and does not allow for much tracking of students' progress or internship opportunities close to where the students are based or where their relatives are located. Costs are also high because of the extensive paper used.

Overall, the absence of a centralized digital internship management system results in inefficiency, poor communication, low data accuracy, delayed processes, and weak monitoring of internship activities.

Major Problems Identified

- The process of placement of internships is manual and takes a considerable amount of time for both students and the placement coordinator.
- There is no overall internet platform that can handle the internship, causing everything to be unorganized.
- Those students who cannot participate in the internship program can also register, as there is no proper validation process.
- The process of assessment and marking is done manually and is therefore slow and prone to errors.
- It is hard to monitor the progress of students when they are placed in companies.
- Students have limited information regarding internship placements around their homes or relatives.
- There are communication delays between students, coordinators, and firms. Much paper is involved in the whole process, making it quite expensive and difficult to record and trace.

1.3 Project Objectives

1.3.1 General Objective

The general objective of this project is to suggest the **design and development of the web-based Internship Service Portal** that seeks to automate the entire internship management process conducted within the College of Electrical Engineering & Computing.

1.3.2 Specific Objectives

The specific objectives of this project are to:

- The whole process of selection and application for an internship should be done online to avoid many manual processes that students go through.
- The system makes it easier for applicants with qualifications to look for and apply for open internship opportunities.
- The student automatically qualifies for the intern program according to the academic and department standards.
- Reduce paperwork and other costs that result from the manual process currently used.
- Improve the ease of access provided to internship coordinators for the purpose of assigning students to companies and their supervisor-managers without stress.

- Store all student information as well as company and internship information in one interface.
- Automate the grading process and send grades over to the registrar's office.
- That would ameliorate the process of communication and clarify the process at large for all parties.
- Provide a better opportunity for the students to select companies for their internship projects that are closer to their residential area, if possible.

1.4 Stakeholders

The major stakeholders involved in the Internship Service Portal are:

1.4.1 Students

- Apply for internships online instead of going forth and back from the department.
- Check the available companies and see their placement status.
- Upload internship reports and follow up on their grades.

1.4.2 Internship Coordinators

- Check and manage student applications
- Match students with companies and supervisors
- Keep an eye on student progress during the internship
- Confirm student eligibility and make reports

1.4.3 Companies

- Share internship opportunities with students
- Accept or reject student applications through the system
- Assess and give feedback on student performance

1.4.4 College of Electrical Engineering and Computing

- Supervise the overall internship process
- Make sure academic rules and requirements are followed
- View reports and internship-related statistics

1.5 Project Scope

Currently, the project is targeted at the **Software Engineering (SE)** and **Computer Science and Engineering (CSE)** departments alone in the College of Electrical Engineering and Computing. The primary intention is to make the process of doing internships much smoother for the students, coordinators, and the organization.

So with this system, students can apply for an internship, view the companies they've been placed in, request supervision, and view grades. Coordinators can process applications, assign students to the companies, and view the progress of everyone. They can also view the reviews from the companies. Companies can create an internship posting and provide reviews on students.

On the flip side, this project will not entail the creation of internships for the other departments, and neither does the management of salaries or stipends. It's all about keeping things simple and within the SE and CSE students.

CHAPTER TWO: REQUIREMENTS ELICITATION REPORT

2.1 Stakeholder List

These are some of the main actors who are actively participating in the current internship:

- **Students:** They do the internship, submit letters and reports, and check their grades.
- **Department:** Issues internship letters, accepts acceptance letters from students, assigns supervisors and companies, organizes presentations, evaluates students, and sends grades to the registrar.
- **Companies:** Accept internship letters, supervise students during the internship, and give numeric grades to the department through supervisors.
- **Registrar:** Sends student data to departments on paper and uploads grades to the e-student portal.

2.2 Interviews

We did semi-structured interviews to understand how things actually work. Here's who we talked to:

- **Students:** 4–6 students to see how they apply, submit letters/reports, and check grades.
- **Department staff:** 1 staff member to understand assigning supervisors, evaluating students, and sending grades.
- **Registrar staff:** 1 person to see how student data is delivered and grades are managed.
- **Companies:** No interviews were conducted.

2.3 Sample Interview Questions

The interviews were semi-structured. The participants included 4–6 students, 1 department staff member, and 1 registrar staff member. No company staff were interviewed.

For Students:

- How do you currently receive internship letters?
- What steps do you follow to apply to a company?
- How do you submit acceptance letters and internship reports?
- How do you view your grades?

For Department Staff:

- How do you assign supervisors and companies for students?
- How do you evaluate student performance?
- How do you send grades to the registrar?

For Registrar Staff:

- How is student data sent to departments before internship?
- How do you process and upload grades to the e-student portal?

2.4 Research on Existing Platforms

We analyzed some existing platforms like Handshake to help design our system related to internships. These platform allow students to search and apply for internships, and in some cases, even track their status. These give feedback from firms and allow students to apply to multiple positions simultaneously.

After that, we also asked students, staff, and department heads about how internships work at our college right now.

From the information gathered and observations, we realized the following:

- ✧ **Student Perceptions:** The most frustrating part for them is generating letters by hand, the process of submitting reports, and checking their grades. The need is for them to view internships and apply for and track them online rather than running around for everything.
- ✧ **Coordinators:** The process of assigning students to the companies and supervisors involves a lot of time, especially when done manually. They are also unable to track reports and feedback from the different students and companies.
- ✧ At the **department** level the challenge is how they manage the paperwork, checking eligibility, and submitting grades to the registrar correctly. There is no one place where everything can be viewed, and hence it is time-consuming and error-prone.
- ✧ **Companies:** The existing platforms like Handshake is helpful for posting general intern ads, but they do not consider the college-specific regulations and criteria for eligibility and grading.

From this, we got a much better idea of what, from the research, our system should go ahead and do:

- Let students apply for internships easily and in a more organized fashion.
- Allow coordinators to assign students quickly and monitor their progress.
- Centralize reports and grade management for the department
- Automatic eligibility checking of students considering college rules
- Track feedback and evaluations in a manner that suits our college workflow.

Therefore, after analyzing the existing platforms and having a discussion with the students as well as administrative staff members, we were able to identify the flaws in the existing system. This helped us understand that what needs to be created in our system so that the process of internship occurs in a smoother and clearer way.

2.5 Overview of Requirements Elicitation

So, to solve our question of what our internship site should provide, we used a combination of methods to collect information:

- ✧ **Observation:** What we essentially observed was how things can be done effectively in the current state. Students go around collecting internship letters, submitting the letters to the firms, and turning in the reports. Departments assign supervisors, review the reports and presentations, and then transmit grades to the registrar. The registrar finally puts grades online on the website for students. Nothing much is electronic, except when the students check their grades online
- ✧ **Interviews:** We conducted some interviews with students, members of the department, and members of the registrar's staff to see how they do the actual internship and what challenges they are facing.
- ✧ **Document Analysis:** To understand the workflow and identify the type of details required and which details need automation in the new system, we analyzed the old letters written for the internship and the report form.
- ✧ **Analytical Insights into Current Platforms:** Apart from the above, we also analyzed platform such as 'Handshake' to understand how they enabled applications, monitored applications, and received feedback from the platform for the students. These platform helped us realize the kind of features we could implement, as adjusted to our college guidelines and structures.

With everything we have considered, we have got a good idea of the things that are lacking, the things that are slow, and the things that the portal must offer to facilitate the internship for the students, coordinators, and departments.

2.6 Summary of Elicitation Findings

Based on our readings, interviews, and observations, we have observed the following regarding the process of internship arrangements:

Who is involved:

Students, Department, Companies, and the Registrar

Student process: They must obtain internship letters from the students, submit them to the companies, and later send the acceptance letters back to the department. They also attend the internship program, complete the reports or projects assigned, do the presentations, and lastly verify their marks on the e-student website once they are uploaded from the registrar.

Department process: The department searches for companies and writes letters for internships, gathers acceptance letters from students, assigns supervisors and companies when necessary, arranges for presentations, rates the students, and manually submits grades to the registrar.

Registrar process: All the student information, such as name, ID, activation and grades, is handled on paper by the registrar; then he uploads grades to the e-student portal once he receives them from the department.

Company process: Companies accept the internship letters, supervise the students in their internship, and provide numeric grades to the department through the supervisors.

Internship duration & evaluation: All students take their internships during a fixed summer period. Evaluation is based on company grades in combination with departmental evaluation of reports and presentations.

Analogous to manual versus digital: Everything in this workflow is almost manual: students, departments, and the registrar handle papers at nearly every step. The only part that's digital is the checking of grades by students on the e-student portal. Companies interact indirectly through students and supervisors rather than making use of any system directly.

Takeaway

We have gained really good insight into the internship process, including the gaps, through the interviews. Currently, everything is slow, really paper-filled, and kind of inefficient. It is clear to

us now that we really need to implement a technology platform, like a portal, to streamline the internship application, management, and grading processes for everyone involved.

Chapter 3: Requirements Analysis and Modeling

3.1 Functional Requirements

These are, basically, the things that the system should do, or what the user can do with the system:

Organize by user

Students:

- Can register themselves and get logging in the portal for all internship features.
- They can search for available internships using filters like company, location, or field.
- Can apply for internships online and get confirmation that the application went through, without having to run around with papers.
- The system itself checks their eligibility based on department, year, and academic status.
- Can directly upload internship reports, projects, or designs on the portal.
- Can request supervision if they need guidance during the internship.
- Can view grades after the department and registrar have submitted them.

Coordinators:

- Can assign students to companies and supervisors quickly through the portal.
- Can track which students applied, submitted reports, and how they're progressing during the internship.
- Can view uploaded reports and projects in one place for easy evaluation.
- Can evaluate students based on reports, presentations, and feedback from companies.
- Can securely submit grades to the registrar.

Companies:

- Can accept internship applications.
- Can supervise students doing internship.
- Can provide grades and feedback to the coordinators or via the portal.

Registrar:

- Can accept grades from the coordinators.
- Can upload grades to the e-student portal so students can view them online.

Organize by function

✧ Student Registration and Login

- Students are able to register and login into the portal where they get all the details about the internship.

✧ Browse and Search Internships

- Students are able to view all the intern positions that are available according to their department and preferred location.

✧ Apply for Internships

- The students can apply online and obtain a notification that the process is complete. There is no need for the students to present the documents to the institutions.

✧ Auto Eligibility Check

- It will automatically verify whether the student qualifies or not, depending on the department, year, and status of the student.

✧ Submit Internship Reports

- Students can upload their reports, projects, or designs straight to the portal.

✧ Provide Students with Company and Supervisor Assignment

- Allocating the students to the companies as well as the supervisors by the coordinators through the portal is very easy.

✧ Monitor Student Progress

- Coordinators can monitor student applications, reports, and internship status using one platform.

✧ Evaluation & Grading

- The departments will have the ability to evaluate students on the basis of reports, presentations, and feedback from the companies.

3.2 Non-Functional Requirements

These are pretty much the attributes the portal should display in other word how the portal should feel and function:

- ✧ **Easy to Use:** The site must be easy to understand and not confusing to students, coordinators, or staff.
- ✧ **Secure:** Only those people will have access to their own data. Nobody else should have access to it.
- ✧ **Fast and Smooth:** The system must always run with speed, regardless of the number of users connected to the system at the time.
- ✧ **Stable:** There should not be a freeze or crash when the program is running.
- ✧ **Accessible Anywhere:** The students have to be able to login to the portal from anywhere.
- ✧ **Easy to Update:** The coordinators and admins should be able to update, alter, or add any information like companies, students, or grades easily.

3.3 System Models (UML Diagrams)

To understand how the portal works, we use **UML diagrams**. These are basically visual ways to show **who does what** and **how the system behaves**.

3.3.1 Use Case Diagram

System: Internship Service Portal

✧ Below there is a list of actors and what their role in the portal plus how they interact with the interface :

1. Student: Registration & Application

- **Step 1 (Verification):** Student enters their **School ID (e.g., UGR/35977/26)**. The portal sends a request to the **Registrar's database** to verify the student is in the 3rd or 4th year.
- **Step 2 (Browsing):** Once logged in, the student views the "Company" table to see active internship posts.
- **Step 3 (Applying):** Student clicks "Apply." The system creates a new entry in the **Application** table, linking the **UGR ID** to the **Internship ID**.
- **Step 4 (Reporting):** After the internship starts, the student uploads their PDF report directly to their application record.

2. Internship Coordinator: Management & Grading

- **Step 1 (Company Approval):** The Coordinator reviews new **Company** sign-ups. They change the company status from "Pending" to "Approved" so they can post internships.
- **Step 2 (Matching):** They monitor the **Application** table to ensure students are being picked by companies.
- **Step 3 (Grading):** After the student uploads a report, the Coordinator reviews it and enters a grade into the **Grade Record**.
- **Step 4 (Finalizing):** The Coordinator clicks "Submit to Registrar," which triggers the automatic sync.

3. Company: Posting & Selection

- **Step 1 (Access):** Once approved by the Coordinator, the Company logs in and adds a new record to the **Internship** table.
- **Step 2 (Review):** The Company views the **Application** table. They see a list of **UGR IDs** and student names who applied.
- **Step 3 (Decision):** The Company changes the Application Status to "Accepted" or "Rejected." This automatically notifies the student portal.

4. Registrar: Data Verification & Grade Collection

- **Step 1 (Authentication):** When a student tries to log in, the Registrar's portal provides the "Status" (Year/Dept) to the internship portal.

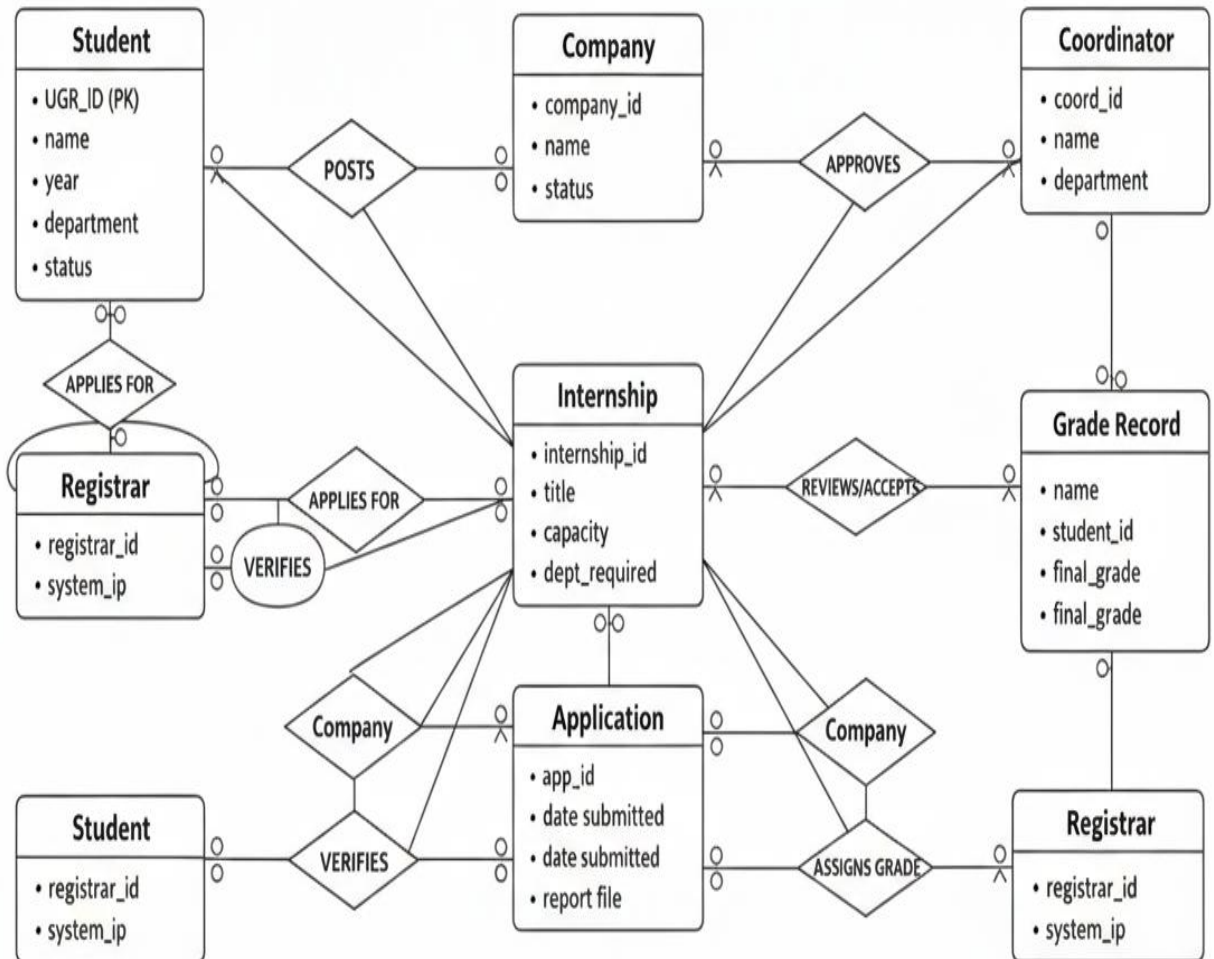
- **Step 2 (Automatic Update):** Once the Coordinator submits a grade, the Registrar's portal receives a secure packet: [UGR_ID: UGR/35977/16 | Grade: A | Course: Internship]. The Registrar's database updates the student's official transcript automatically.

There relationships among each other:

- They directly interact with the system in application procedures, report uploads, supervision requests, and viewing grades.
- Coordinators collaborate to assign, monitor, grade and submit.
- Companies cooperate by giving feedback and grades.
- Registrar interacts by receiving grades and updating the portal.

Purpose of this diagram:

- To graphically depict how each actor interacts with the portal and what activity they can conduct.
- Allows the tracking of activity flows and different roles assigned to users in the process of managing internships.



3.3.2 Use Case Descriptions

Use case descriptions give an idea of what kinds of users are using the Internship Service Portal and what they can do with it. It tells about what actions students, coordinators, companies, and the registrar will carry out in terms of significant actions, what must happen before a particular action, and what must happen to a client after an event occurs. Overall, it acts as a guide in understanding what a user views in a system. It is very helpful in ensuring that every required task in a system is covered in a thorough manner and makes it easier to test it.

Through understanding these "user stories," it is possible to comprehend accurately what makes a correct intern work flow in this portal, what makes a smooth flow of communication in this context, and what makes a good intern-seeking experience in an efficient and effective manner.

From this diagram, we can easily tell how responsibilities are shared between users and how the system acts as a central platform which connects everybody. Students apply for internships, send reports, ask for supervision, and query grades within the system. Coordinators manage the applications, assign companies and supervisors, monitor progress, and grade students. Companies interact with the system by posting internships and providing feedback, while the registrar handles grade verification and publishing.

The use case diagram summarizes the whole workflow in a very simple and graphic form. It replaces the old manual, paper-based process with a structured digital flow, avoids potential confusion among different stakeholders, and makes sure that every important interaction is supported by the system. This diagram acts as a blueprint for guiding system design, development, and validation.

Chapter 4: Project Presentation and Validation

4.1 Requirements Validation Process

The validation procedure allows everything that we had designed to be true in the context of the Internship Service Portal is in fact true. This means that whether everything that is proposed in the form of features will help to provide a solution to the problems being faced by the students, coordinators, firms, and the registrar is being cross-checked here. We have tested these requirements to ensure that these are in fact pragmatic and feasible in real life scenarios that we can measure. This involved validating all the requirements to be free from ambiguity and relevance to the solution proposed by the portal.

Validation Techniques Used

Specifically concerning the validation of requirements, we basically sought validation from people who will actually implement the system. The plan has been shown to students, coordinators, and department representatives, as well as the registrar, to check whether they found it sensible and addressed their needs.

We also researched existing platforms such as Handshake to understand how they perform their tasks. This is crucial to ensure that our portal provides practical functionalities and functions as intended.

Lastly, we put our plan into practice with some real-world internship examples such as applying for an internship position, submitting reports, and allocating supervisors so that we can ensure that all aspects of the current plan actually work properly.

4.2 Feedback and Approvals from Stakeholders

Now that we had all those requirements and features in place, it was time to sit down with the key individuals who are ultimately going to be using our portal, and that includes students, internship coordinators, department staff, as well as the registrar at each university. We explained all our intentions to them and asked them to review it to see if it made any sense to them as well as addressed all their needs and is feasible in real life too. We provided them with feedback of all they liked and disliked in it and made all those corrections before they gave us their approval to go ahead with it.

Sample Comments by Stakeholders

Name: Anonymous

- “It would be great if we can view all the openings for the internship program at once instead of asking around.”
- “It has got to be an automatic check of eligibility. we waste time calculating that ourselves.”
- “The uploading of reports online would have saved so much time instead of handing papers around.
- "It would be cool if we could track our application and be able to see what's going on."

From: Coordinators / Department Staff

- "Manual allocation of students is taking too long. We need an allocation method that can be done inside the system."
- "Tracking reports and presentations is messy; a dashboard could solve this problem."
- “There should be notification when students apply and when they submit reports.”

Registrar / Admin:

- "It is just too much to upload grades manually when it can be automatically uploaded through the portal."
- "Much better to have centralized data like fewer pieces of paper and fewer errors."

CHAPTER FIVE: REQUIREMENT ENGINEERING TOOLS USED

5.1 Overview

This chapter describes the tools used during the requirements engineering activities of the University Internship Portal System project. The tools supported documentation, requirement analysis, note-taking during elicitation, and system modeling. Each tool was selected based on availability, simplicity, and suitability for the project scope.

5.2 Description of Each Tool

5.2.1 Microsoft Word

Microsoft Word was used as the primary documentation tool throughout the project. It was used to prepare formal project documents such as the requirements elicitation report, system descriptions, and other written deliverables required for the Software Requirements Engineering course. The tool supported structured writing, formatting, and easy editing.

5.2.2 Notepad

Notepad was used during the requirements elicitation phase to take quick notes, especially during interviews when pen and paper were not available. It provided a simple and fast way to record responses from stakeholders before organizing them into formal documentation.

5.2.3 Gemini AI

Gemini AI was used to support research and analysis activities. It helped in understanding concepts related to requirements engineering, clarifying system processes, and supporting analytical thinking during the project. The tool was used as a reference and analysis aid, not as a replacement for stakeholder input.

5.2.4 Draw.io

Draw.io was used for system modeling purposes. It was used to create:

- Use case diagrams
- Entity Relationship Diagrams (ERD)

These diagrams helped visualize system actors, interactions, and data relationships in a clear and understandable manner.

5.3 Tools Used Throughout the Project

The following tools were used during different phases of the project:

Tool	Purpose
Microsoft Word	Project documentation
Notepad	Interview note-taking
Gemini AI	Research and analysis
Draw.io	Use case and ERD diagram

5.4 Justification for the Choice of the Tools

The software used was chosen due to the following reasons:

- ✧ **Microsoft Word** is preferred as it is widely available and capable of producing formal documentation.
- ✧ **Notepad** was chosen because of its simplicity along with its rapid availability.
- ✧ **Gemini AI** was employed for helping in research and understanding of analytical concepts related to requirement engineering.
- ✧ We chose **draw.io** because it is very easy to use and is an efficient tool in creating system diagrams.

All the tools short-listed were adequate enough to support the activity of requirements engineering in the project without making things complicated.